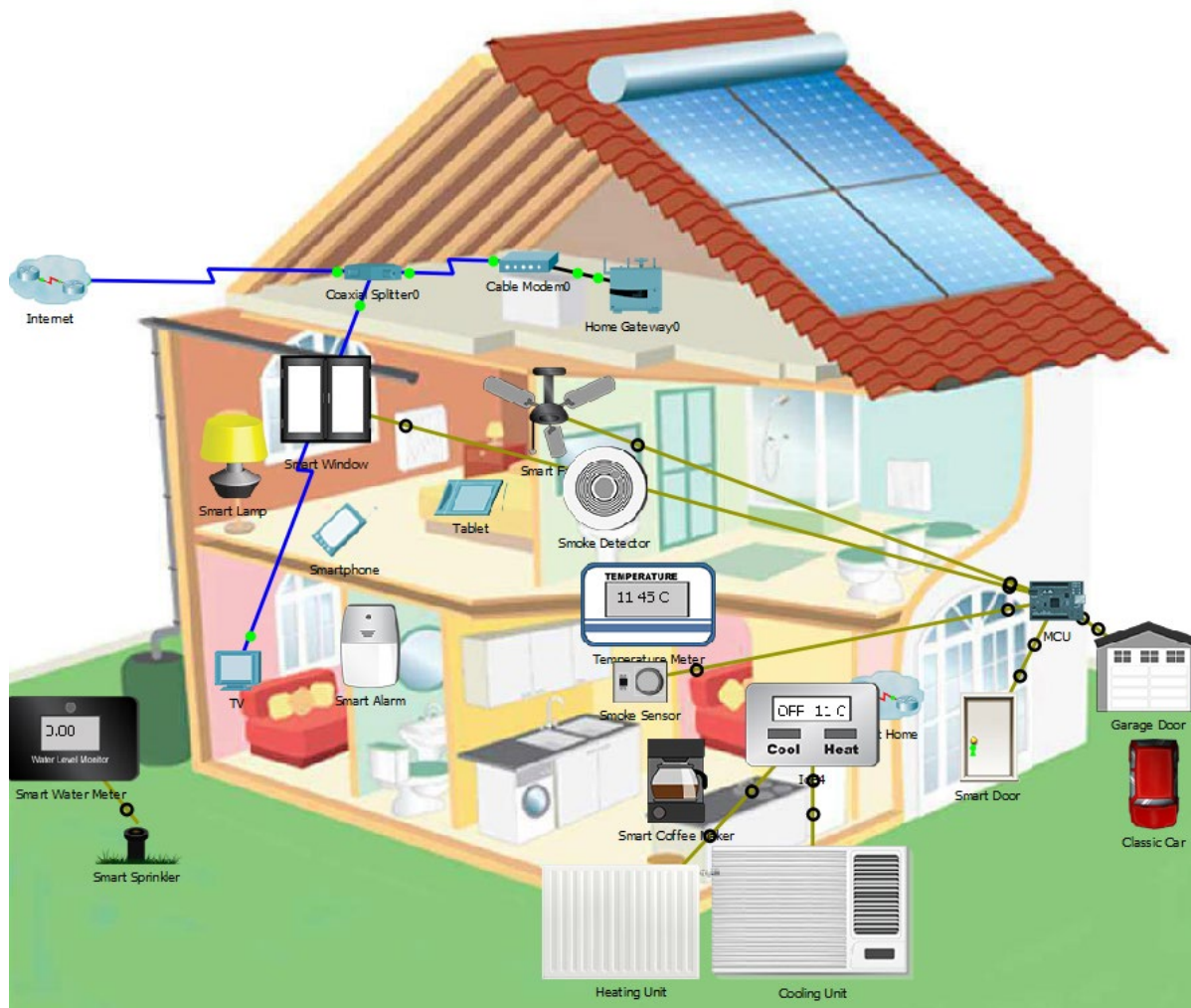


Packet Tracer – Threat Modeling at the Device Layer

Topology

The topology is a home IoT system that has been prototyped in Packet Tracer. It shows a cutaway view of a home with different sensors, actuators, and connections shown.



Objectives

In this Packet Tracer, you will begin the threat modeling process for the device layer of the IoT attack surface.

Part 1: Identifying the Security Objectives

Part 2: Exploring and Diagraming the Physical Network

Part 3: Creating an Inventory of Assets of the Physical Device Attack Surface

Part 4: Identifying Potential Threats with the STRIDE Model

Background / Scenario

A home owner has been reading about the IoT and is very enthusiastic about the capabilities of home automation IoT systems. The home owner wants to install such a system in his house but does not know how to do so. He has contacted a company that can design and install the system. The company focuses on security throughout the system design, provisioning, and development process. They are currently developing a threat model for the system. You have been hired by the company and your first task is to complete the threat model.

The house is 3500 sq. ft. and includes two stories and an attic. The customers are away from home regularly and have requested the safest house possible. The customer wants to be able to remotely monitor the house and wants the following IoT-supported systems:

- Climate control
- Smoke / fire
- Temperature issues out of the normal range
- Door and window locks
- Lawn watering
- Local alarm and emergency department messaging

The system should be controllable locally and through the cloud. The user should be able to access the controller from a web browser from inside the network as well as remotely through a smartphone app. This will allow the customers to monitor or control the system when they are away.

The system should collect and store data from the remote sensors and various actions should take place based on the input from those sensors. For example, if the temperature goes above the maximum range, it probably means the AC is not working and someone needs to be notified ASAP. If the system detects smoke, the local alarm should sound, and the customer and the fire department should be alerted. Data from the system should be retained and analyzed. In addition, the customer should be able to change the threshold values that trigger different actuators and events as necessary, either locally or through the mobile app. The triggers and behaviors, data analytics, and remote-control access are all available through a home automation cloud application service that the system will interface with.

The home owners should have password protected accounts for access to the system. In addition, the company should have access to system diagnostics in case problems occur with the system. Only the homeowner should have access to the cloud applications.

It is also important to make note of these other details of the house:

- 3 bedrooms, 1 den, 2 baths
- 2 stories and an attic
- 1 main front door and 1 side door entry
- 2 sliding doors to the backyard – one coming from the master bedroom
- 2-car garage

The team has designed the system and it is your job to perform threat modeling on the design.

You will start by creating a threat model for system. The home automation system has been prototyped in Packet Tracer. The system is very similar to the home IoT system that you explored earlier in the course. Because the threat modeling process is very detailed, we will be breaking it up into four linked labs. The first three labs are for each layer of the IoT attack surface. In the final Packet Tracer, you will rate the risks and determine how the risks will be managed.

Required Resources

- Packet Tracer 7.1 or later

Part 1: Identifying the Security Objectives

The first activity in the threat modeling process is to establish the security objectives for the system.

Step 1: Provide objectives for each of the six categories.

There are six categories of security objectives that are often used to define the security needs of a system or organization. Answer the questions to help develop these objectives. Answer the questions from the point of view of both the company and the customer, where appropriate.

Identity

Visualize the application and identify the security objectives of this system based on the requirements.

What access and authorization controls should be in place to document who is accessing the IoT system?

Access to network resources

user name and password ,normal user account

Should there be any machine-to-machine (M2M) access controls in place?

yes ,because M2M :

It is easy to roll out and maintain/It offers higher range/The security and privacy issues in IoT networks are resolved by using it

Device Identity Management

Financial

Document the financial losses that could occur due to a failure of the system, system components, or security breach.

What is the potential financial impact to the customer if components of the system malfunction?

Beyond the immediate financial impact of data loss, other consequences include a loss of customer confidence, corporate liability and the loss of current and future business

unnecessary costs due to excessive energy consumption

theft in the case of open doors or the garage door

If a threat actor was able to gain access to the home network in a security breach, what losses could occur?

access data

Reputation

Document any possible impact on the customer's reputation if the IoT safety/security system is attacked.

What would be the repercussions for the owner if their home-based safety/security system was attacked?

failure of military equipment, and breaches of national security secrets

They can disrupt phone and computer networks or paralyze systems, making data unavailable that can damage his business' reputation and erode the trust his customers have for him. This, in turn, could potentially lead to: loss of customers. loss of sales.

Privacy and Regulation

Document the impact of any privacy concerns as well as regulation requirements for this system.

Are there any privacy concerns for any of the data collected or used by this system?

Generate the data from a temperature sensor in an irrigation system
may not have any privacy or regulatory concerns

Does the owner care that logs are being kept about their access and movement throughout the house (motion sensors)?

yes

Identify any data that could cause privacy concerns for the owner of this system.

As most of the devices may be battery operated, due to less processing power the security and privacy is a major issue in IoT. Authentication, Identification and device heterogeneity are the major security and privacy concerns in IoT

Availability Guarantees

Document the expected availability and guaranteed uptime of the IoT system. Is this system required to be available at all times?

yes

Is there any acceptable downtime that can be tolerated for this system? Explain.

No, the system must be "living

Safety

Document the potential impacts to physical welfare of people and physical damage to equipment and facilities. This is particularly important in industrial control system (ICS) environments.

Part 2: Exploring and Diagraming the Physical Network

Step 1: Open the Packet Tracer Network.

The Packet Tracer network is an interactive demonstration of a home automation IoT network. It is the same network that was used in Chapter 1.

- Open the **Packet Tracer - Threat Modeling at the IoT Device Layer.pka** file.

- b. Follow the instructions to become familiar with some of the functions of the network if you do not know how it works.
- c. Feel free to explore the network further.

Step 2: Diagram the network.

Use the blank floor plans that are included in Appendix A of the PDF version of these Packet Tracer instructions. Diagram the IoT devices on the floor plan. Place the sensors, actuators, and devices at the locations on the floor plan that seem to match the PT network. Label each device.

Part 3: Creating an Inventory of Assets of the Physical Device Attack Surface

You will now complete an inventory of the devices in the IoT network by entering them into a table.

Step 1: List all the assets.

Fill in the table that is provided in **Appendix B** of this Packet Tracer with all the IoT device physical assets that are part of the home automation network. Also add the device role.

What are some roles that devices can take in an IoT network?

they are attached to a particular object that operates through the internet, enabling the transfer of data among objects or people automatically without human intervention

Step 2: Determine the interactions between assets.

Continue to work in the table from Step 1. Fill-in the relationships between the devices in the **Works With** column. Which sensors and actuators work together?

Device	Device Role	Works With
Part 1: Physical Device		
Smoke control	Sensor	home Gateway/wifi
Smart door	Controler	wifi
Smart coffee maker	Actuator	wifi/Smart Battery
TV	Actuator	Coaxial splitter
Smart Lamp	Actuator	wifi/Battery

Device	Device Role	Works With
Garage Door	Sensor/Actuator	wifi
smart Solar panel	Sensor	Gateway/Battery
Smart Fan	Sensor/Actuator	Wifi
Smart Door	Actuator	wifi
Smart Solar Panel	Sensor and Actuator	Gateway/Battery
Smart Sprinkle	Actuator	Smart Watter Meter
Water Meter	Sensor/Actuator	Smart Sprinkle
Temperature Gauge	Sensor	wifi
Smoke Detector	Sensor and actuator	wifi
Smart alarm	Actuator	wifi
Smart Window	Actuator	TV/Coaxial Spliter

Part 4: Identifying Potential Threats with the STRIDE Model

In this part, you will identify threats using the [STRIDE methodology](#). Try to describe as many threats as possible based on your experience in the course, the [OWASP IoT vulnerabilities page](#), and other information sources.

Use the STRIDE model to create a list of potential threats

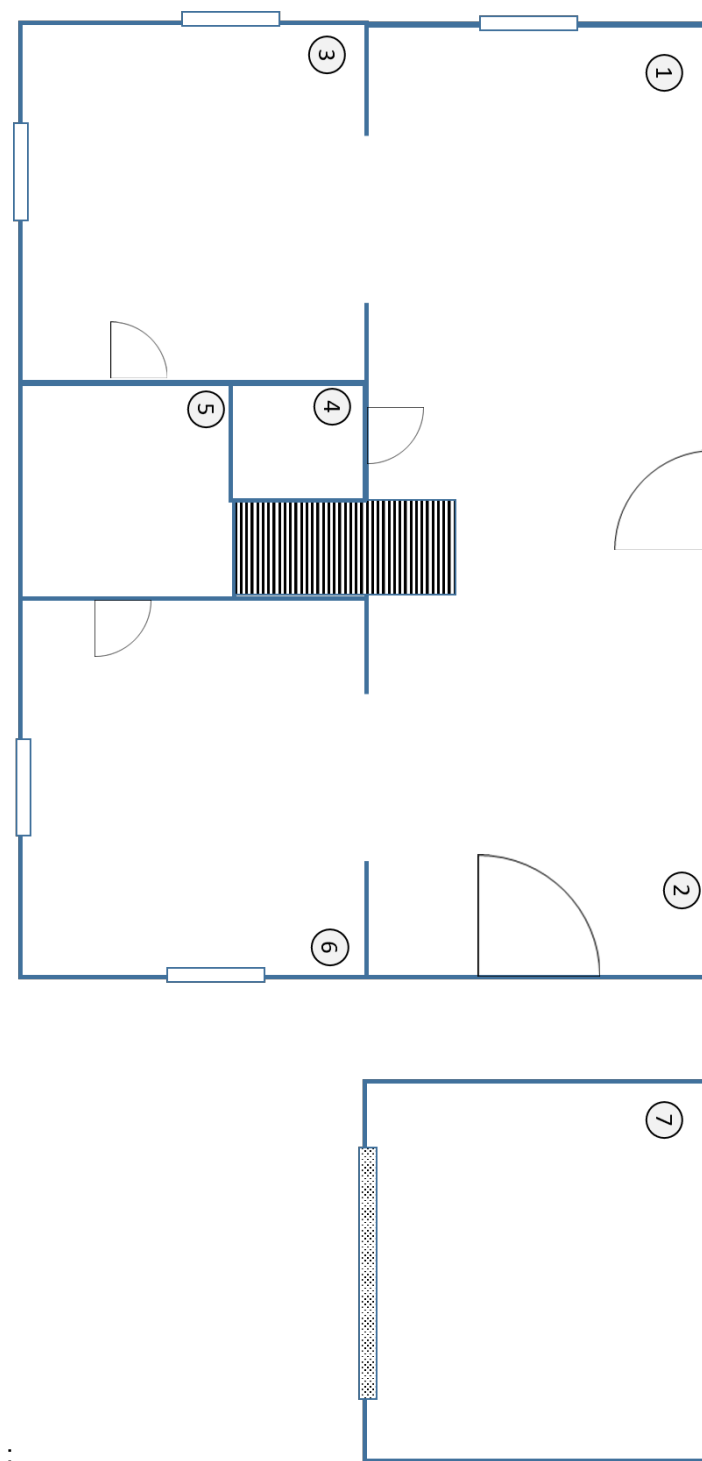
Complete this table with threats for each category in the STRIDE threat model. Add potential threats that could occur for each STRIDE category. Include the type of threat using the OWASP terminology where possible.

Threat type	Asset type	Threats
(S)poofing – can an attacker pretend to be someone he is not, or falsify data?	Sensors	

Threat type	Asset type	Threats
	Actuators	
(T)ampering – can an attacker successfully inject falsified data into the system?	Sensors	
	Actuators	
(R)epudiation – can a user pretend that a transaction did not happen?	Sensors	
	Actuators	
(I)nformation Disclosure – can the device leak confidential data to unauthorized parties?	Sensors	
	Actuators	
(D)enial of Service – can the device be shut down or made unavailable maliciously?	Sensors	
	Actuators	
(E)scalation of Privilege – can users get access to privileged resources meant only for admins or superusers?	Sensors	
	Actuators	

Appendix A: Floor Plans

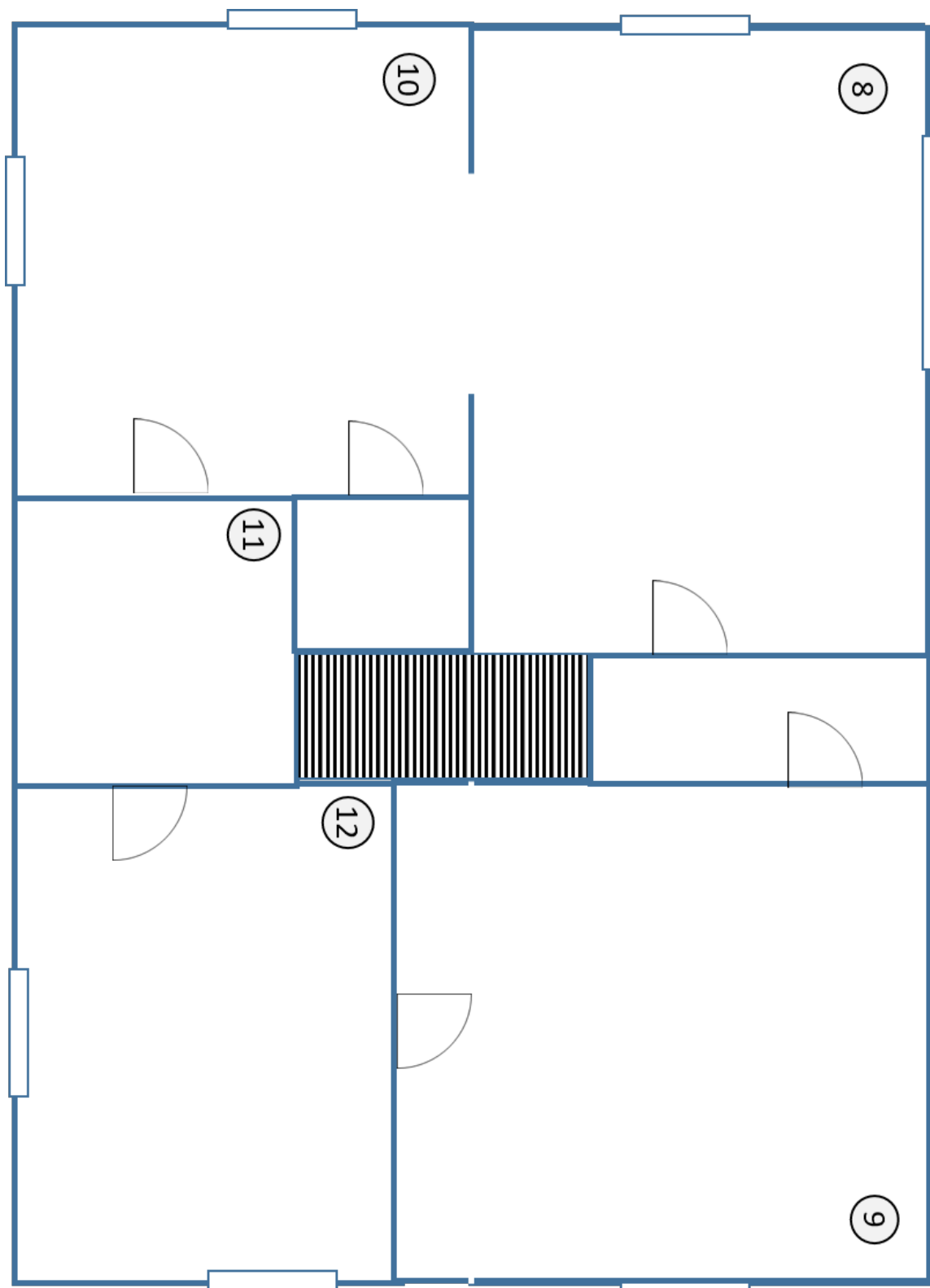
Ground floor



Ground Floor Legend:

Number	Room
1	living room
2	dining room
3	TV room
4	closet
5	bathroom
6	kitchen
7	garage

Upper Floor:



Upper Floor Legend:

Number	Room
8	bedroom
9	bedroom
10	bedroom
11	closet
12	bathroom

Appendix B: Asset Table

Device	Device Role	Works With...
Part 1: Physical Device		

Appendix C: STRIDE Table

Threat type	Asset type	Threats
(S)poofing – can an attacker pretend to be someone he's not or falsify data?	Sensors	
	Actuators	
(T)ampering – can an attacker successfully inject tampered data into the system?	Sensors	
	Actuators	
(R)epudiation – can a user pretend that a transaction didn't happen?	Sensors	
	Actuators	
(I)nformation Disclosure – can the application leaking confidential data to unauthorized parties?	Sensors	
	Actuators	
(D)enial of Service – can the device be shut down or made unavailable maliciously?	Sensors	
	Actuators	
(E)scalation of Privilege – can users get access to privileged resources meant only for admins or superusers?	Sensors	
	Actuators	